Cancer Genomic

The Impact of Non-Synonymous SNPs on Protein Function and Disease Susceptibility in Breast Cancer: A Comparative Human Genomic Analysis

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INTRODUCTION

Study of the totality of DNA sequence and gene expression differences between tumor cells and normal host cells It involves using molecular and genomic profiling to study the causes of cancer

Cancer is a genetic disease caused by accumulation of DNA mutations and epigenetic alterations leading to unrestrained cell proliferation and neoplasm formation.

Cancer Genomics helps us to:

- Discover the environmental factors that cause mutations and identify mutation that cause mutation
- Understand the molecular mechanisms that cause mutations

Research Limitations

- Heterogeneity of cancer genomes
- Ethical concerns related to genetic testing and privacy
- Analyzing and interpreting vast amounts of genomic data accurately and efficiently

Experimental Applications

- Cancer Research
- Precision Medicine

- Pharmacogenomics

- Public Health and Epidemiology

- Genetic Counseling

- Drug Discovery and Development

Project Workflow

1. Data acquisition and pre-processing

2. DNA Sequence alignment

3. Genome annotation

4. Comparative genome analysis of SNPs

5. Identification of genetic markers

6. Comparative genome analysis

Results

1. Data acquisition and pre-processing

I. Collection of Data: Collect WGS of Human Breast Cancer Data from NCBI

Data Retrieval

A total of 7 single whole genome sequence of clinical metagenomics samples associated with a Breast cancer in human with the bio project accession number (PRJNA1086561) were retrieved from the National Center for Biotechnology Information, NCBI (https://www.ncbi.nlm.nih.gov/). All the samples were collected and registered by the Navin Lab (Navin).

Table: General Sequence Properties

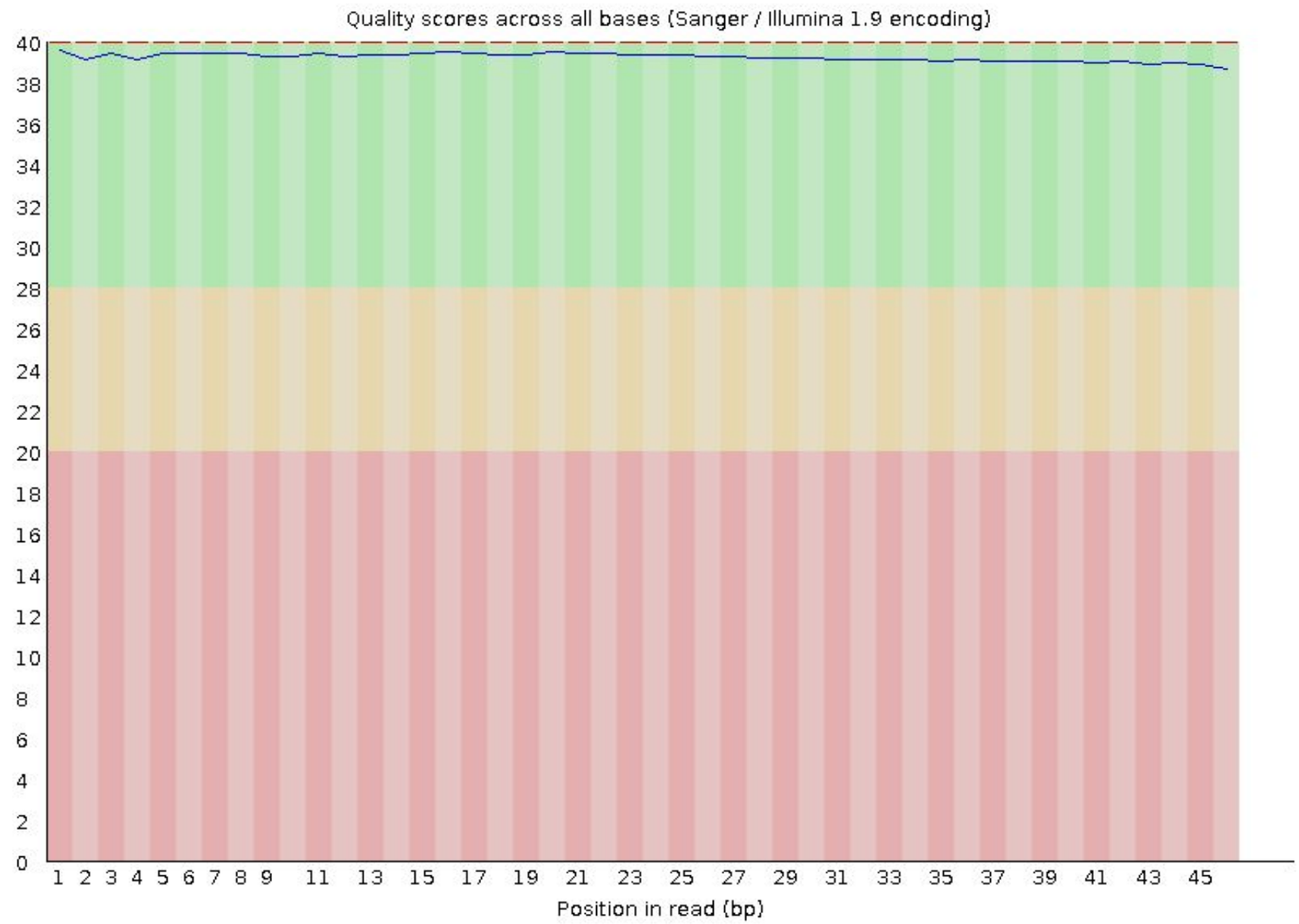
Bio project	Bio sample	SRA	*Sample ID*	Size(MB)	Region	Isolation source	Submitter	Year	Sequencing platform
PRJNA1086561	SAMN48409197	SRR33501133	ECIS57T_chip2	3MB	USA		Navin Lab (Navin)	2025	ILLUMINA (Illumina HiSeq 2500)
PRJNA1086561	SAMN48409184	SRR33505761	ECIS57T_chip2	4.9Mb	USA			2025	
PRJNA1086561	SAMN48409184	SRR33501178	Cellline_mixing_experiment_1_2	6.9Mb	USA			2022	
PRJNA1086561	SAMN48409184	SRR33505745	Cellline_mixing_experiment_1_2	7.6Mb	USA			2022	
PRJNA1086561	SAMN48409184	SRR33505733	Cellline_mixing_experiment_1_2	11.2Mb	USA			2022	
PRJNA1086561	SAMN48409184	SRR33505775	Cellline_mixing_experiment_1_2;	122.4Gb	USA			2022	
PRJNA1086561	SAMN48409184	SRR33505757	Cellline_mixing_experiment_1_2	**8Mb**	USA			2025	

1. Data acquisition and pre-processing

ii. Quality Control and Genome Assembly:

Quality control:

Assess data quality, perform trimming, and filter out low-quality reads from FASTQ using FASTQC to ensure reliable results



1. Data acquisition and pre-processing

ii. Quality Control and Genome Assembly:

Genome Assembly:

- Download SRA sample data in FASTQ format using Galaxy.
- Assemble the paired-end reads using the SPAdes tool.
- Obtain contig and scaffold files from SPAdes results.
- Evaluate genome quality and completeness using QUAST :



2. DNA Sequence Alignment

Mapping to Reference

- Whole-genome sequencing (WGS) reads were obtained from a breast cancer sample (SRR33505761).
- The reads were aligned to the human reference genome using **Geneious Mapper v9.1.8**.
- The alignment focused specifically on **chromosome 19**.
- A **low-sensitivity setting** was used to balance speed and accuracy.
- The process produced **contigs** for further analysis.

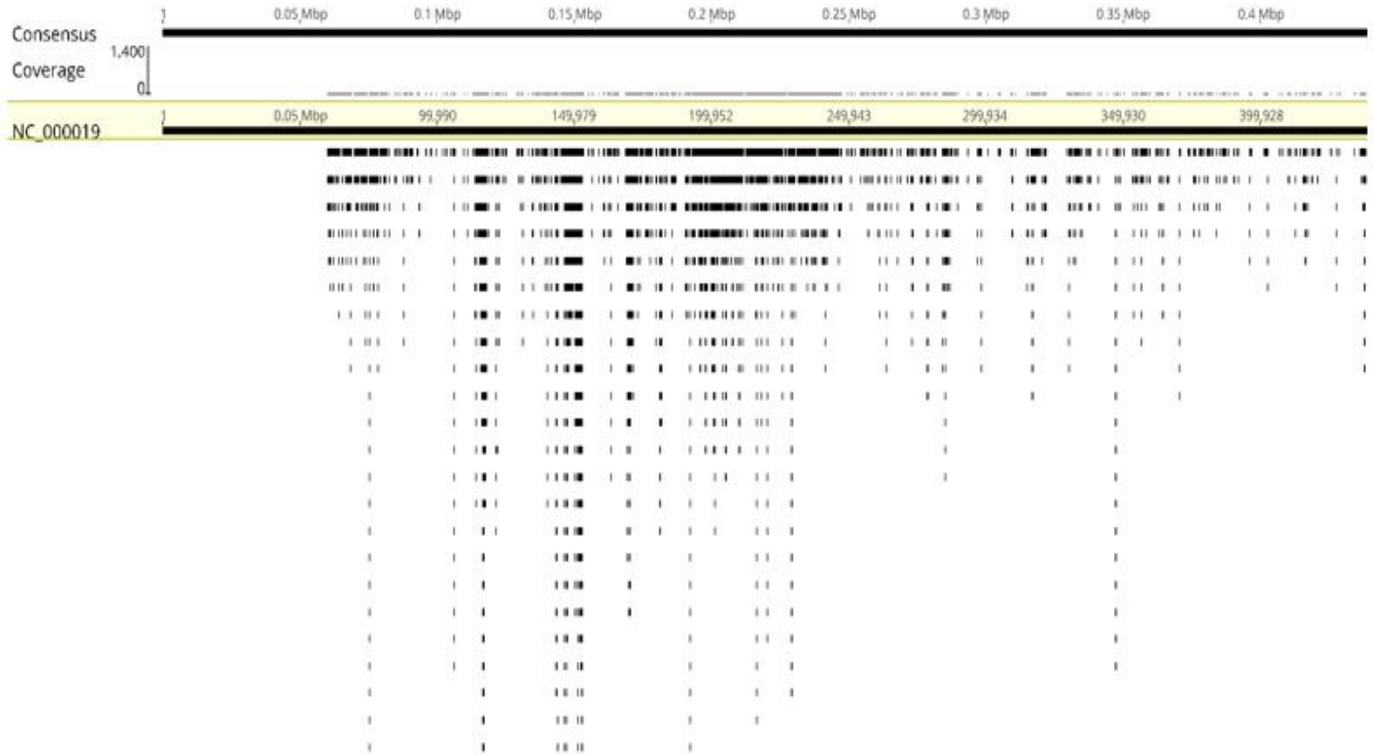


Figure: Aligned reads from a Breast cancer dataset (mapped to human Chromosome 19.

3. Genome Annotation

Functional Annotation

- **Sample:** Annotate SRR33505761 (Breast Cancer Genome) to the **Reference:** Human Chromosome 19
- **Tool:** Reference-guided annotation
- **Identified Features:** CDS, Genes, RNA elements
- **Detected Variants:** CNVs, SVs
- **Predicted Regions:** Promoters and regulatory elements
- **Outcome:** Functional insights into gene dysregulation and cancer progression

Table 1: Summary of Annotated Genomic Features from SRA Run Accession SRR33505761 Aligned to Human Chromosome 19

Chromosomal Features	SRR33505761
CDS	7,522
Centromere	1
Exon	221
gene	2,641
Mature Peptide	477
Misc Feature	22,871
Misc_Recomb	2
Misc-RNA	761
Mobile Element	7
mRNA	7,522
ncRNA	1,098
Precursor _RNA	142
Protein _Bind	92
Regulatory	7,071
Repeat Region	6
Signal peptide	621
source	1
Transit _peptide	42
tRNA	(7)

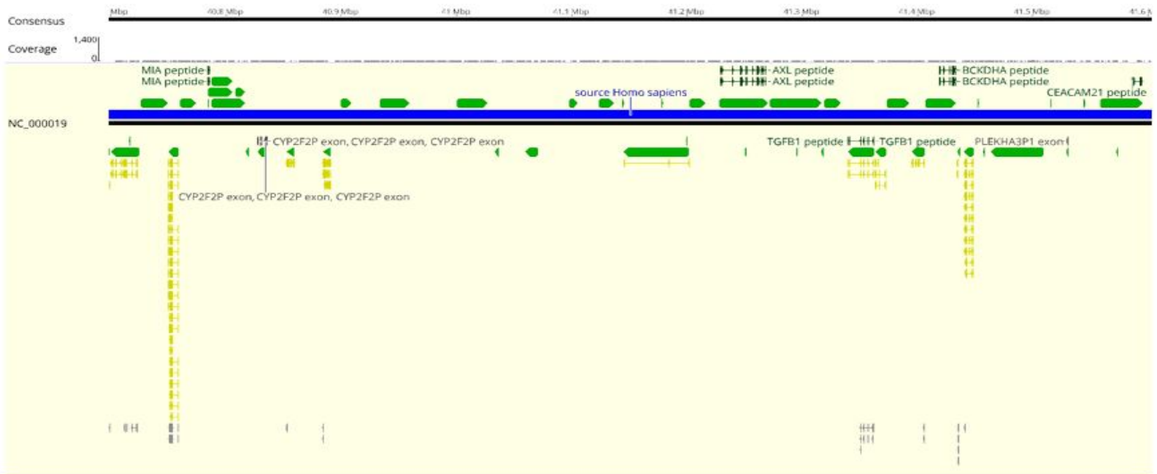


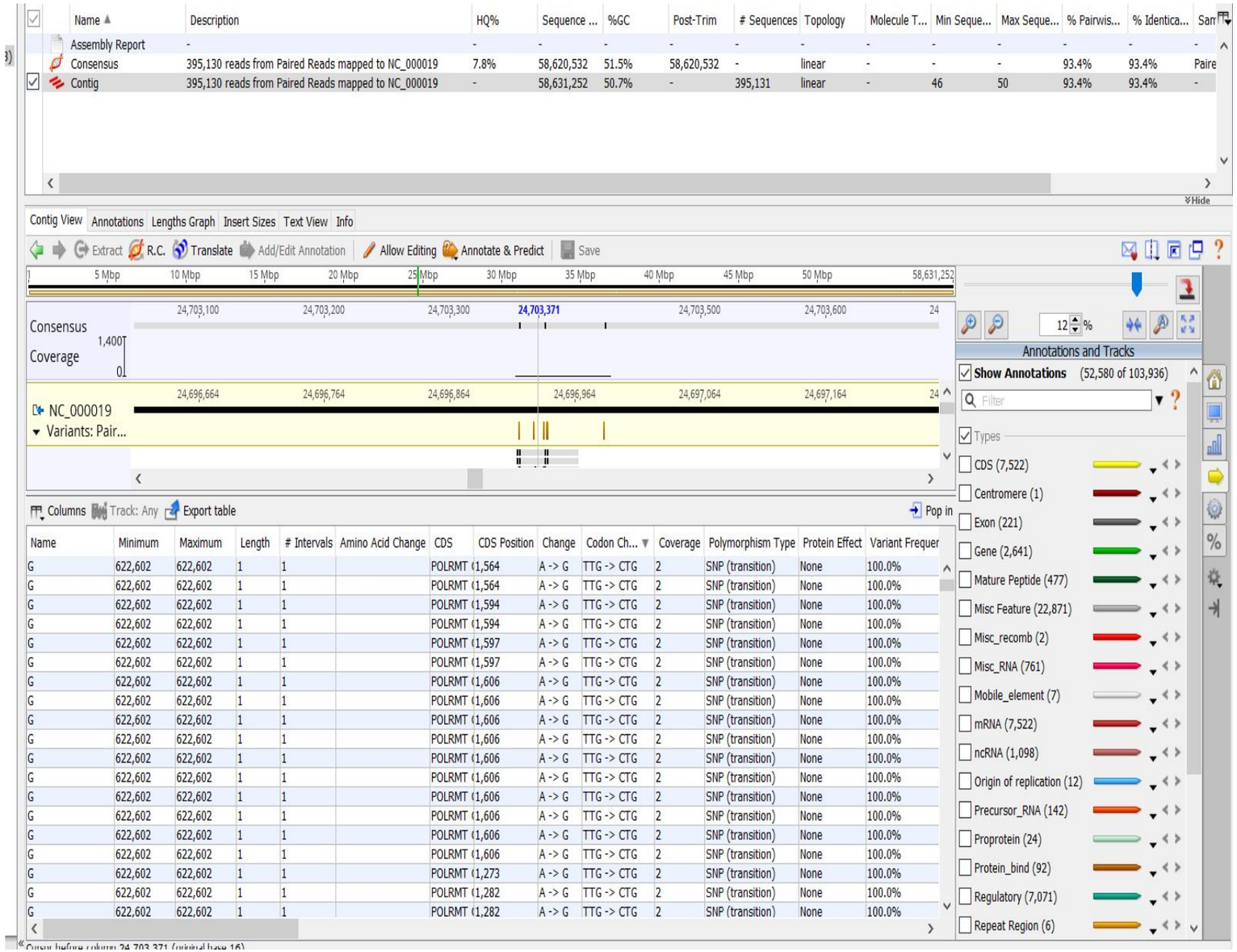
Figure 2: Representative Genomic View of an Annotated Breast Cancer Sample SRR33505761Aligned **to Human Chromosome 19.**

4. Comprehensive Genome Analysis of SNPs

i. Variant Calling Analysis and Effect Prediction

- **Variant calling** done in **Geneious Mapper v9.0.5** using parameters: frequency ≥ 0.25, P ≤ 10^{-5}, strand-bias ≤ 65%.
- **Reads** aligned to **human chromosome 19** to detect **SNPs** and **InDels**.
- **Variants** analyzed for **missense** and **splice site** effects in 100-kb intervals

- Identified **non-synonymous SNPs** as potential **biomarkers** for **breast cancer therapy**

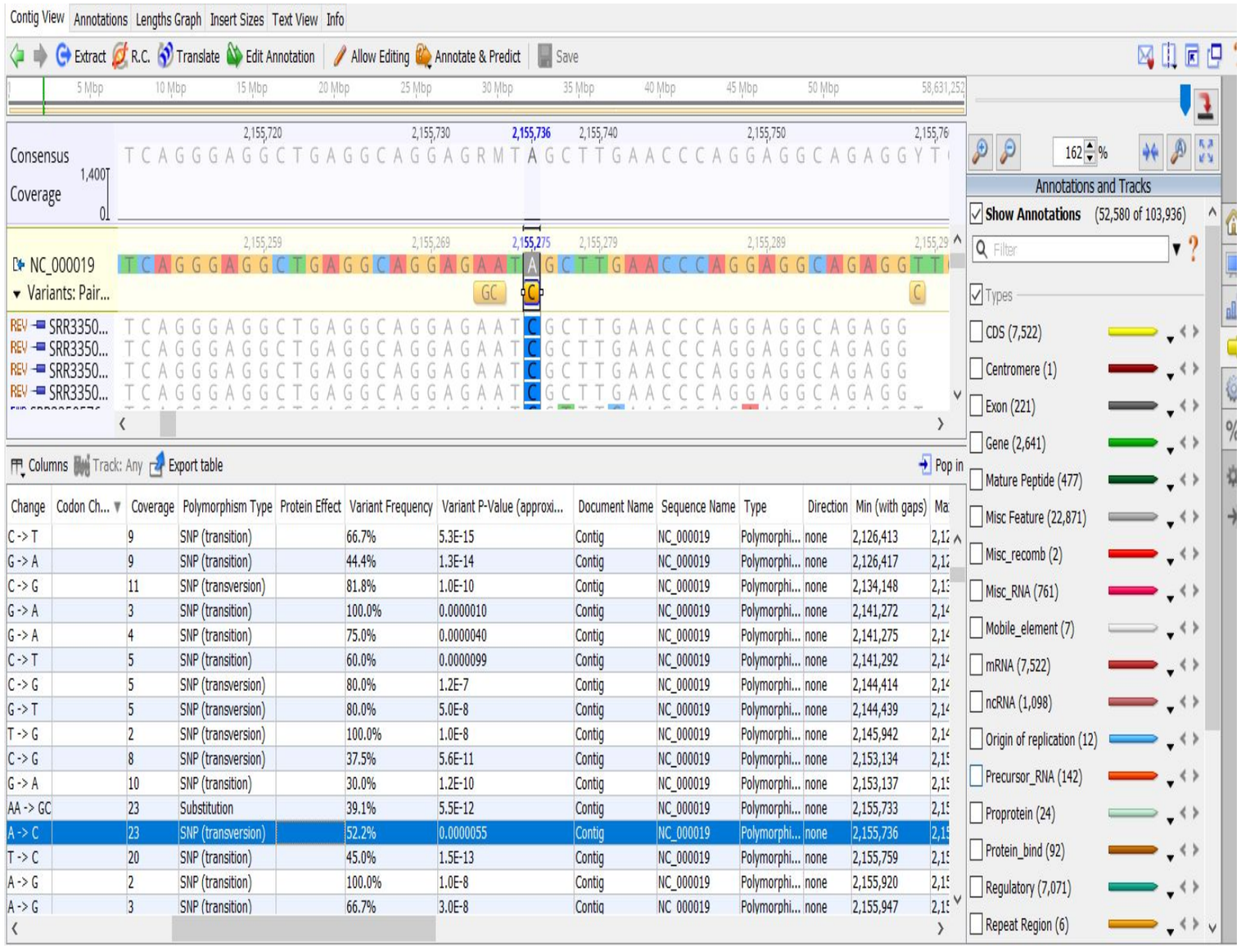


4. Comprehensive Genome Analysis of SNPs

ii. Variant calling Analysis

Interpretation of Variant Calling Results:

- Reads aligned to reference genome to find variations.
- True SNPs confirmed with multiple read support.
- Variants classified as SNPs, insertions, or deletions.
- Affected genes or regions identified for relevance.
- Functional effects predicted on protein or gene level.
- Data exported to Excel for further analysis.
- **Purpose:** Understand biological and disease significance of variants

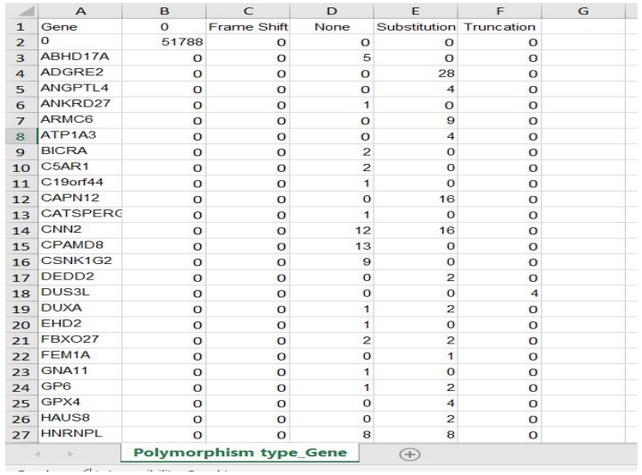


Change	Codon Ch...	Coverage	Polymorphism Type	Protein Effect	Variant Frequency	Variant P-Value (approxi...	Document Name	Sequence Name	Type	Direction	Min (with gaps)
C -> T		9	SNP (transition)		66.7%	5.3E-15	Contig	NC_000019	Polymorphi...	none	2,126,413
G -> A		9	SNP (transition)		44.4%	1.3E-14	Contig	NC_000019	Polymorphi...	none	2,126,417
C -> G		11	SNP (transversion)		81.8%	1.0E-10	Contig	NC_000019	Polymorphi...	none	2,134,148
G -> A		3	SNP (transition)		100.0%	0.0000010	Contig	NC_000019	Polymorphi...	none	2,141,272
G -> A		4	SNP (transition)		75.0%	0.0000040	Contig	NC_000019	Polymorphi...	none	2,141,275
C -> T		5	SNP (transition)		60.0%	0.0000099	Contig	NC_000019	Polymorphi...	none	2,141,292
C -> G		5	SNP (transversion)		80.0%	1.2E-7	Contig	NC_000019	Polymorphi...	none	2,144,414
G -> T		5	SNP (transversion)		80.0%	5.0E-8	Contig	NC_000019	Polymorphi...	none	2,144,439
T -> G		2	SNP (transversion)		100.0%	1.0E-8	Contig	NC_000019	Polymorphi...	none	2,145,942
C -> G		8	SNP (transversion)		37.5%	5.6E-11	Contig	NC_000019	Polymorphi...	none	2,153,134
G -> A		10	SNP (transition)		30.0%	1.2E-10	Contig	NC_000019	Polymorphi...	none	2,153,137
AA -> GC		23	Substitution		39.1%	5.5E-12	Contig	NC_000019	Polymorphi...	none	2,155,733
A -> C		23	SNP (transversion)		52.2%	0.0000055	Contig	NC_000019	Polymorphi...	none	2,155,736
T -> C		20	SNP (transition)		45.0%	1.5E-13	Contig	NC_000019	Polymorphi...	none	2,155,759
A -> G		2	SNP (transition)		100.0%	1.0E-8	Contig	NC_000019	Polymorphi...	none	2,155,920
A -> G		3	SNP (transition)		66.7%	3.0E-8	Contig	NC_000019	Polymorphi...	none	2,155,947

4. Comprehensive Genome Analysis

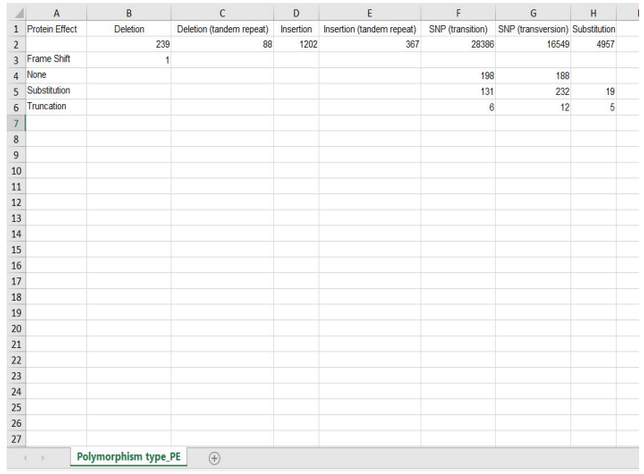
iii. Variant Data Cleaning

- **Data cleaning :**removes errors to improve data quality and accuracy
- Tool: Tableau software is used to clean, prepare, and visualize data for accurate analysis.
- **In single datasets**, polymorphism types were mapped to genes and protein effects and protein effect to gene
- **In merged datasets**, mapping sample IDs to various attributes like gene, polymorphism type, protein effect, and more.



Gene	0	Frame Shift	None	Substitution	Truncation
0	51788	0	0	0	0
ABHD17A	0	0	5	0	0
ADGRE2	0	0	0	28	0
ANGPTL4	0	0	0	4	0
ANKRD27	0	0	1	0	0
ARMC6	0	0	0	9	0
ATP1A3	0	0	0	4	0
BICRA	0	0	2	0	0
C5AR1	0	0	2	0	0
C19orf44	0	0	1	0	0
CAPN12	0	0	0	16	0
CATSPERG	0	0	1	0	0
CNN2	0	0	12	16	0
CPAMD8	0	0	13	0	0
CSNK1G2	0	0	9	0	0
DEDD2	0	0	0	2	0
DUS3L	0	0	0	0	4
DUXA	0	0	1	2	0
EHD2	0	0	1	0	0
FBXO27	0	0	2	2	0
FEM1A	0	0	0	1	0
GNA11	0	0	1	0	0
GP6	0	0	1	2	0
GPX4	0	0	0	4	0
HAUS8	0	0	0	2	0
HNRNPL	0	0	8	8	0

Polymorphism type_Gene



Protein Effect	Deletion	Deletion (tandem repeat)	Insertion	Insertion (tandem repeat)	SNP (transition)	SNP (transversion)	Substitution
	239	88	1202	367	28386	16549	4957
Frame Shift	1						
None					198	188	
Substitution					131	232	19
Truncation					6	12	5

Polymorphism type_PE

Gene	0	Frame Shift	None	77:2225VH7I	Substitution	Truncation
0	51788	0	0	0	0	0
ABHD17A	0	0	5	0	0	0
ADGRE2	0	0	0	0	28	0
ANGPTL4	0	0	0	0	4	0
ANKRD27	0	0	1	0	0	0
ARMC6	0	0	0	0	9	0
ATP1A3	0	0	0	0	4	0
BICRA	0	0	2	0	0	0
C5AR1	0	0	2	0	0	0
C19orf44	0	0	1	0	0	0
CAPN12	0	0	0	0	16	0
CATSPERG	0	0	1	0	0	0
CNN2	0	0	12	0	16	0
CPAMD8	0	0	13	0	0	0
CSNK1G2	0	0	9	0	0	0
DEDD2	0	0	0	0	2	0
DUS3L	0	0	0	0	0	4
DUXA	0	0	1	0	2	0
EHD2	0	0	1	0	0	0
FBXO27	0	0	2	0	2	0
FEM1A	0	0	0	0	1	0
GNA11	0	0	1	0	0	0
GP6	0	0	1	0	2	0
GPX4	0	0	0	0	4	0
HAUS8	0	0	0	0	2	0
HNRNPL	0	0	8	0	8	0

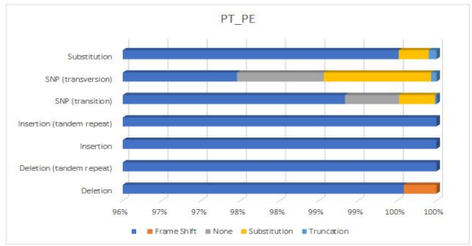
Variant Data Cleaning Results (Single Dataset)

4. Comprehensive Genome Analysis

iv. visualization

- Visualization was last and final step.
- Visualization done using **ClustVis** (web-based clustering tool, BETA)
- Included **single** and **merged** dataset results
- Mapped **polymorphism type → gene → protein effect**
- Explored **relationships between gene variants and functional impact**





Thank You